

HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		award (1) for each correct formula A H_2 B NaNO_3 C $\text{Zn}(\text{NO}_3)_2$ D H_2SO_4		4		4		
		(ii)		filter (1) wash (with water) and dry (to constant mass) (1) accept wash (with water) and evaporate to dryness	2			2		2
	(b)	(i)		$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ award (1) for correct reactants award (1) for correct products ignore incorrect balancing		2		2		
		(ii)	I	add an indicator (to alkali) (1) add acid (dropwise) until indicator / it changes colour (1) accept named indicator, ignore incorrect colour change answers not linked so award (1) for 'add acid until it changes colour'	2			2		2

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					AO1	AO2	AO3	Total	Maths	Prac
			II	award (1) for either of following add recorded acid volume (to 25.0 cm ³ alkali) add same volume of acid and alkali as used in stage 1 without indicator (1)	2			2		2
			III	award (1) for any reference to evaporation of water e.g. allow solution to evaporate to dryness leave to evaporate until dry evaporate the water heat to dryness	1			1		1
				Question 3 total	7	6	0	13	0	7